

Question	Marking Guidance	Mark	Comments
09.1	Iron ions 1. Haemoglobin binds/associates with oxygen OR Haemoglobin transports/loads oxygen; Sodium ions 2. <u>Co-transport</u> of glucose/amino acids (into cells); 3. (Because) sodium moved out by active transport/Na – K pump; 4. Creates a sodium concentration/diffusion gradient; 5. Affects osmosis/water potential; Phosphate ions 6. Affects osmosis/water potential; 7. Joins nucleotides/in phosphodiester bond/in backbone of DNA/RNA/in nucleotides; 8. Used in/to produce ATP; 9. Phosphorylates other compounds (usually) making them more reactive; 10. Hydrophilic/water soluble part of phospholipid bilayer/membrane;	5 max	Must have MP1 for 5 max 3 max for sodium and 3 max for phosphate 1. Ignore reference to 2⁺ or 3⁺ in Fe²⁺ or Fe³⁺ Accept 5. OR 6. – not both Accept for 1 mark, Sodium ions cause water reabsorption in kidneys OR Sodium ions establish resting potential (in neurones) OR Sodium ion diffusion creates action potential 8. Reject 'energy produced'

09.2	1. Phospholipid (bilayer) allows movement/diffusion of non-polar/lipid-soluble substances; 2. Phospholipid (bilayer) prevents movement/diffusion of polar/charged/lipid-insoluble substances OR (Membrane) proteins allow polar/charged substances to cross the membrane/bilayer; 3. Carrier proteins allow active transport; 4. Channel/carrier proteins allow facilitated diffusion/co-transport; 5. Shape/charge of channel / carrier determines which substances move; 6. Number of channels/carriers determines how much movement; 7. Membrane surface area determines how much diffusion/movement; 8. Cholesterol affects fluidity/rigidity/permeability;	5 max	1. and 2. Accept correct named examples 1 and 2 Ignore water 1. Accept phospholipid (bilayer) allows movement/diffusion of O ₂ /CO ₂ 1. Accept water-insoluble 2. Accept water-soluble 4. Accept aquaporins allow osmosis 6 and 7 accept correct reference to faster/slower/rate for 'how much movement' 7. Accept microvilli / Golgi (apparatus) / ER / rER 7. Accept surface area to volume for 'surface area' 8. Accept cholesterol affects vesicle formation/ endocytosis/exocytosis/phagocytosis;
TOTAL		10	